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AC Induction Motor Fundamentals

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INTRODUCTION

AC induction motors are the most common motors used in industrial motion control systems, as well as in main powered home appliances. Simple and rugged design, low-cost, low maintenance and direct connection to an AC power source are the main advantages of AC induction motors.

Various types of AC induction motors are available in the market. Different motors are suitable for different applications. Although AC induction motors are easier to design than DC motors, the speed and the torque control in various types of AC induction motors require a greater understanding of the design and the characteristics of these motors.

This application note discusses the basics of an AC induction motor; the different types, their characteristics, the selection criteria for different applications and basic control techniques.

BASIC CONSTRUCTION AND OPERATING PRINCIPLE

Like most motors, an AC induction motor has a fixed outer portion, called the stator and a rotor that spins inside with a carefully engineered air gap between the two.

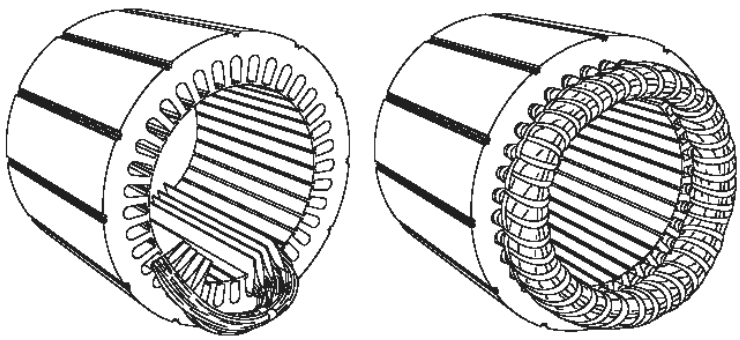
Virtually all electrical motors use magnetic field rotation to spin their rotors. A three-phase AC induction motor is the only type where the rotating magnetic field is created naturally in the stator because of the nature of the supply. DC motors depend either on mechanical or electronic commutation to create rotating magnetic fields. A single-phase AC induction motor depends on extra electrical components to produce this rotating magnetic field.

Two sets of electromagnets are formed inside any motor. In an AC induction motor, one set of electromagnets is formed in the stator because of the AC supply connected to the stator windings. The alternating nature of the supply voltage induces an Electromagnetic Force (EMF) in the rotor (just like the voltage is induced in the transformer secondary) as per Lenz's law, thus generating another set of electromagnets; hence the name – induction motor. Interaction between the magnetic field of these electromagnets generates twisting force, or torque. As a result, the motor rotates in the direction of the resultant torque.

Stator

The stator is made up of several thin laminations of aluminum or cast iron. They are punched and clamped together to form a hollow cylinder (stator core) with slots as shown in Figure 1. Coils of insulated wires are inserted into these slots. Each grouping of coils, together with the core it surrounds, forms an electromagnet (a pair of poles) on the application of AC supply. The number of poles of an AC induction motor depends on the internal connection of the stator windings. The stator windings are connected directly to the power source. Internally they are connected in such a way, that on applying AC supply, a rotating magnetic field is created.

FIGURE 1: A TYPICAL STATOR



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Rotor

The rotor is made up of several thin steel laminations with evenly spaced bars, which are made up of aluminum or copper, along the periphery. In the most popular type of rotor (squirrel cage rotor), these bars are connected at ends mechanically and electrically by the use of rings. Almost 90% of induction motors have squirrel cage rotors. This is because the squirrel cage rotor has a simple and rugged construction. The rotor consists of a cylindrical laminated core with axially placed parallel slots for carrying the conductors. Each slot carries a copper, aluminum, or alloy bar. These rotor bars are permanently short-circuited at both ends by means of the end rings, as shown in Figure 2. This total assembly resembles the look of a squirrel cage, which gives the rotor its name. The rotor slots are not exactly parallel to the shaft. Instead, they are given a skew for two main reasons.

The first reason is to make the motor run quietly by reducing magnetic hum and to decrease slot harmonics.

The second reason is to help reduce the locking tendency of the rotor. The rotor teeth tend to remain locked under the stator teeth due to direct magnetic attraction between the two. This happens when the number of stator teeth are equal to the number of rotor teeth.

The rotor is mounted on the shaft using bearings on each end; one end of the shaft is normally kept longer than the other for driving the load. Some motors may have an accessory shaft on the non-driving end for mounting speed or position sensing devices. Between the stator and the rotor, there exists an air gap, through which due to induction, the energy is transferred from the stator to the rotor. The generated torque forces the rotor and then the load to rotate. Regardless of the type of rotor used, the principle employed for rotation remains the same.

Speed of an Induction Motor

The magnetic field created in the stator rotates at a synchronous speed (N_S).

EQUATION 1:

$$N_S = 120 \times \frac{f}{P}$$

where:

N_S = the synchronous speed of the stator magnetic field in RPM
P = the number of poles on the stator
f = the supply frequency in Hertz

The magnetic field produced in the rotor because of the induced voltage is alternating in nature.

To reduce the relative speed, with respect to the stator, the rotor starts running in the same direction as that of the stator flux and tries to catch up with the rotating flux. However, in practice, the rotor never succeeds in "catching up" to the stator field. The rotor runs slower than the speed of the stator field. This speed is called the Base Speed (N_b).

The difference between N_S and N_b is called the slip. The slip varies with the load. An increase in load will cause the rotor to slow down or increase slip. A decrease in load will cause the rotor to speed up or decrease slip. The slip is expressed as a percentage and can be determined with the following formula:

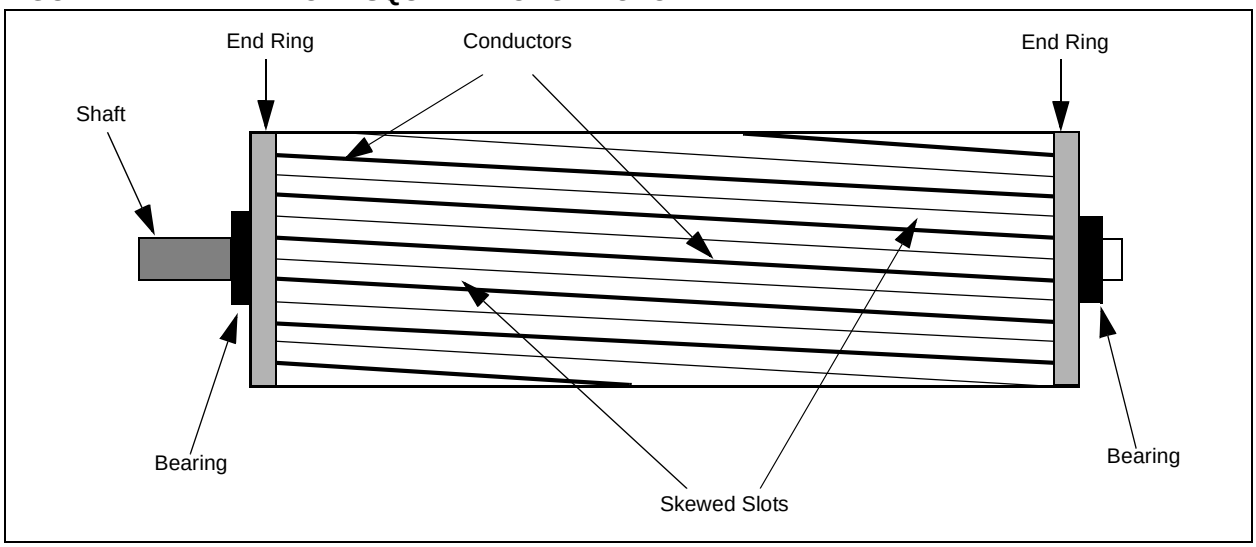
EQUATION 2:

$$\%\ slip = \frac{N_S - N_b}{N_S} x100$$

where:

N_S = the synchronous speed in RPM
N_b = the base speed in RPM

FIGURE 2: A TYPICAL SQUIRREL CAGE ROTOR



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TYPES OF AC INDUCTION MOTORS

Generally, induction motors are categorized based on the number of stator windings. They are:

- Single-phase induction motor
- Three-phase induction motor

Single-Phase Induction Motor

There are probably more single-phase AC induction motors in use today than the total of all the other types put together. It is logical that the least expensive, lowest maintenance type motor should be used most often. The single-phase AC induction motor best fits this description.

As the name suggests, this type of motor has only one stator winding (main winding) and operates with a single-phase power supply. In all single-phase induction motors, the rotor is the squirrel cage type.

The single-phase induction motor is not self-starting. When the motor is connected to a single-phase power supply, the main winding carries an alternating current. This current produces a pulsating magnetic field. Due to induction, the rotor is energized. As the main magnetic field is pulsating, the torque necessary for the motor rotation is not generated. This will cause the rotor to vibrate, but not to rotate. Hence, the single-phase induction motor is required to have a starting mechanism that can provide the starting kick for the motor to rotate.

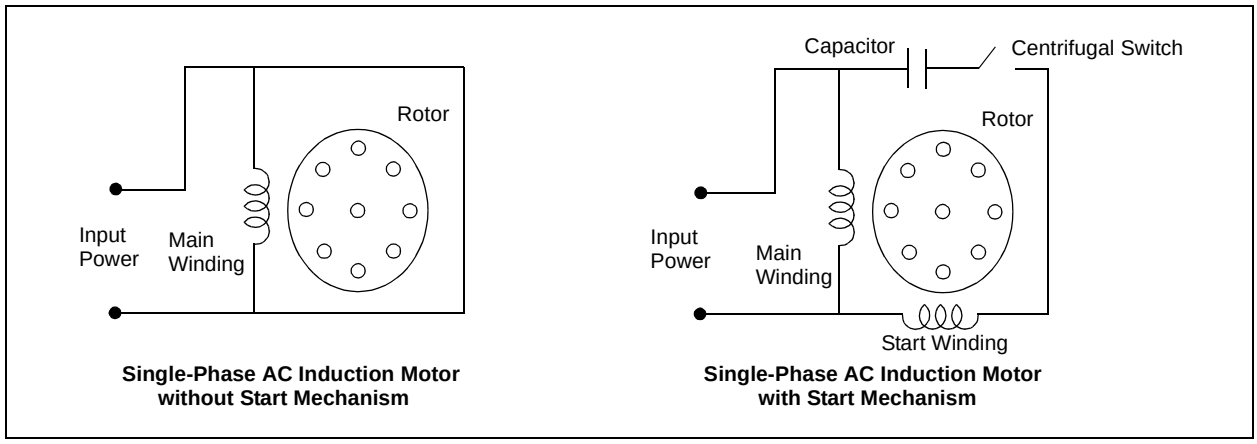
The starting mechanism of the single-phase induction motor is mainly an additional stator winding (start/auxiliary winding) as shown in Figure 3. The start winding can have a series capacitor and/or a centrifugal switch. When the supply voltage is applied, current in the main winding lags the supply voltage due to the main winding impedance. At the same time, current in the start winding leads/lags the supply voltage depending on the starting mechanism impedance. Interaction between magnetic fields generated by the main winding and the starting mechanism generates a resultant magnetic field rotating in one direction. The motor starts rotating in the direction of the resultant magnetic field.

Once the motor reaches about 75% of its rated speed, a centrifugal switch disconnects the start winding. From this point on, the single-phase motor can maintain sufficient torque to operate on its own.

Except for special capacitor start/capacitor run types, all single-phase motors are generally used for applications up to 3/4 hp only.

Depending on the various start techniques, single-phase AC induction motors are further classified as described in the following sections.

FIGURE 3: SINGLE-PHASE AC INDUCTION MOTOR WITH AND WITHOUT A START MECHANISM



Single-Phase AC Induction Motor without Start Mechanism

Single-Phase AC Induction Motor with Start Mechanism

TYPICAL NAME PLATE OF AN AC INDUCTION MOTOR

A typical name plate on an AC induction motor is shown in Figure 19.

FIGURE 19: A TYPICAL NAME PLATE

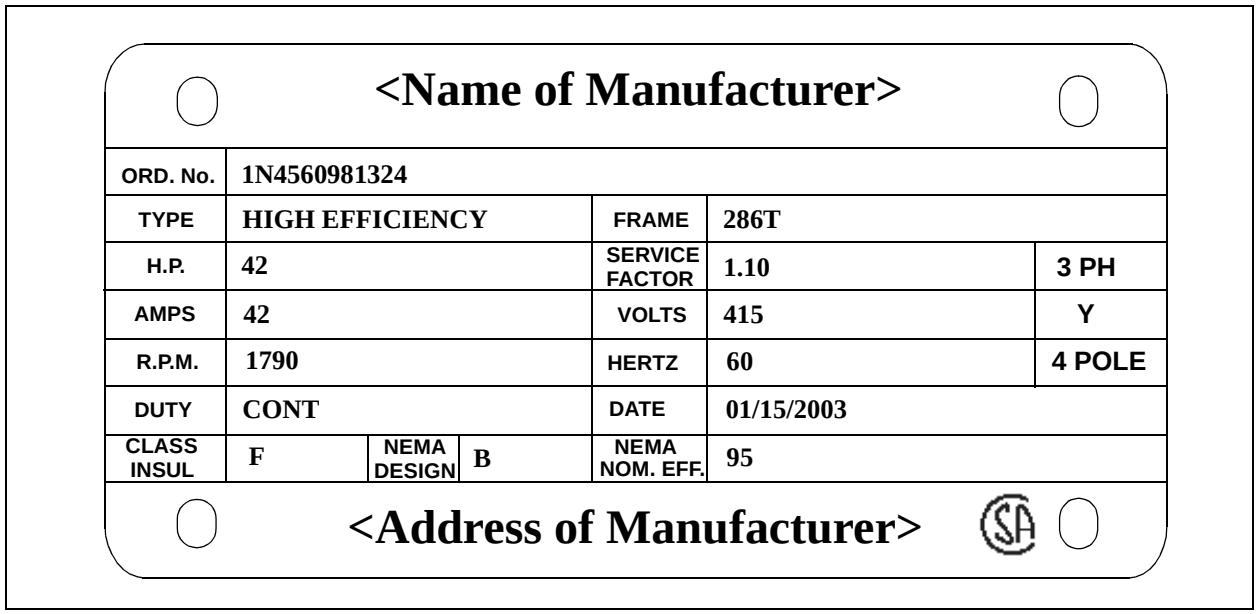


TABLE 2: NAME PLATE TERMS AND THEIR MEANINGS

Term	Description
Volts	Rated terminal supply voltage.
Amps	Rated full-load supply current.
H.P.	Rated motor output.
R.P.M	Rated full-load speed of the motor.
Hertz	Rated supply frequency.
Frame	External physical dimension of the motor based on the NEMA standards.
Duty	Motor load condition, whether it is continuos load, short time, periodic, etc.
Date	Date of manufacturing.
Class Insulation	Insulation class used for the motor construction. This specifies max. limit of the motor winding temperature.
NEMA Design	This specifies to which NEMA design class the motor belongs to.
Service Factor	Factor by which the motor can be overloaded beyond the full load.
NEMA Nom. Efficiency	Motor operating efficiency at full load.
PH	Specifies number of stator phases of the motor.
Pole	Specifies number of poles of the motor.
	Specifies the motor safety standard.
Y	Specifies whether the motor windings are start (Y) connected or delta (Δ) connected.

NEED FOR THE ELECTRICAL DRIVE

Apart from the nonlinear characteristics of the induction motor, there are various issues attached to the driving of the motor. Let us look at them one by one.

Earlier motors tended to be over designed to drive a specific load over its entire range. This resulted in a highly inefficient driving system, as a significant part of the input power was not doing any useful work. Most of the time, the generated motor torque was more than the required load torque.

For the induction motor, the steady state motoring region is restricted from 80% of the rated speed to 100% of the rated speed due to the fixed supply frequency and the number of poles.

When an induction motor starts, it will draw very high inrush current due to the absence of the back EMF at start. This results in higher power loss in the transmission line and also in the rotor, which will eventually heat up and may fail due to insulation failure. The high inrush current may cause the voltage to dip in the supply line, which may affect the performance of other utility equipment connected on the same supply line.

When the motor is operated at a minimum load (i.e., open shaft), the current drawn by the motor is primarily the magnetizing current and is almost purely inductive. As a result, the PF is very low, typically as low as 0.1. When the load is increased, the working current begins to rise. The magnetizing current remains almost constant over the entire operating range, from no load to full load. Hence, with the increase in the load, the PF will improve.

When the motor operates at a PF less than unity, the current drawn by the motor is not sinusoidal in nature. This condition degrades the power quality of the supply line and may affect performances of other utility equipment connected on the same line. The PF is very important as many distribution companies have started imposing penalties on the customer drawing power at a value less than the set limit of the PF. This means the customer is forced to maintain the full-load condition for the entire operating time or else pay penalties for the light load condition.

While operating, it is often necessary to stop the motor quickly and also reverse it. In applications like cranes or hoists, the torque of the drive motor may have to be controlled so that the load does not have any undesirable acceleration (e.g., in the case of lowering of loads under the influence of gravity). The speed and accuracy of stopping or reversing operations improve the productivity of the system and the quality of the product. For the previously mentioned applications, braking is required. Earlier, mechanical brakes were in use. The frictional force between the rotating parts and the brake drums provided the required braking. However, this type of braking is highly inefficient. The heat generated while braking represents loss of energy. Also, mechanical brakes require regular maintenance.

In many applications, the input power is a function of the speed like fan, blower, pump and so on. In these types of loads, the torque is proportional to the square of the speed and the power is proportional to the cube of speed. Variable speed, depending upon the load requirement, provides significant energy saving. A reduction of 20% in the operating speed of the motor from its rated speed will result in an almost 50% reduction in the input power to the motor. This is not possible in a system where the motor is directly connected to the supply line. In many flow control applications, a mechanical throttling device is used to limit the flow. Although this is an effective means of control, it wastes energy because of the high losses and reduces the life of the motor valve due to generated heat.

When the supply line is delivering the power at a PF less than unity, the motor draws current rich in harmonics. This results in higher rotor loss affecting the motor life. The torque generated by the motor will be pulsating in nature due to harmonics. At high speed, the pulsating torque frequency is large enough to be filtered out by the motor impedance. But at low speed, the pulsating torque results in the motor speed pulsation. This results in jerky motion and affects the bearings' life.

The supply line may experience a surge or sag due to the operation of other equipment on the same line. If the motor is not protected from such conditions, it will be subjected to higher stress than designed for, which ultimately may lead to its premature failure.

All of the previously mentioned problems, faced by both consumers and the industry, strongly advocated the need for an intelligent motor control.

With the advancement of solid state device technology (BJT, MOSFET, IGBT, SCR, etc.) and IC fabrication technology, which gave rise to high-speed microcontrollers capable of executing real-time complex algorithm to give excellent dynamic performance of the AC induction motor, the electrical Variable Frequency Drive became popular.

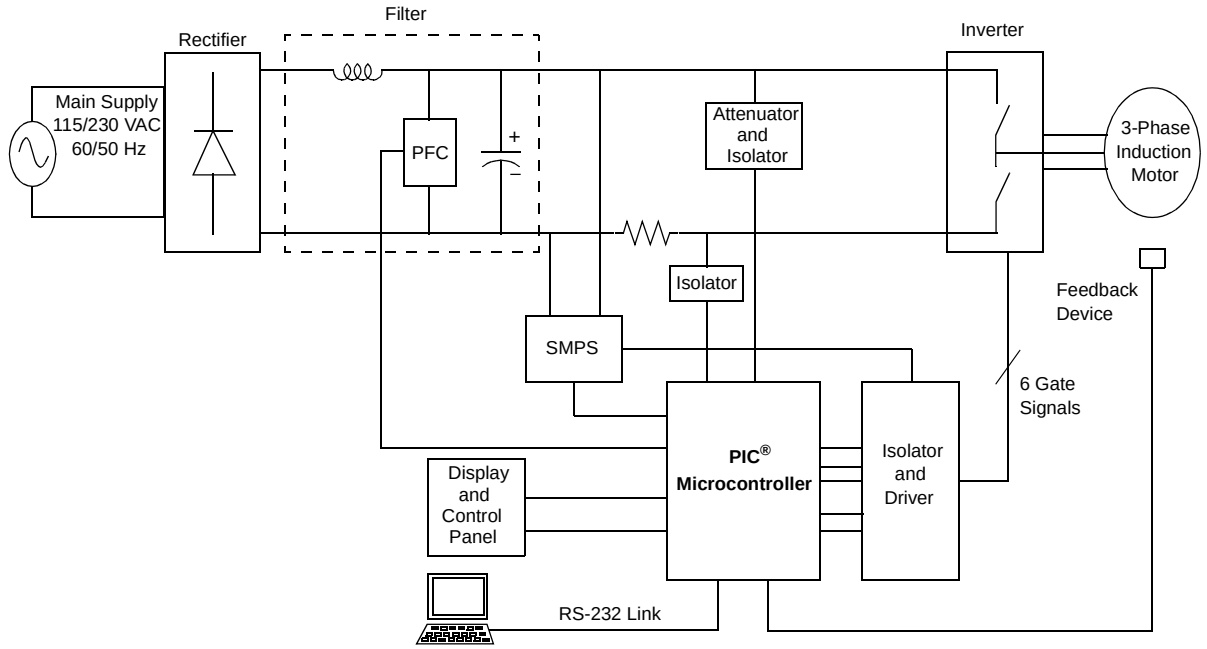
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VARIABLE FREQUENCY DRIVE (VFD)

The VFD is a system made up of active/passive power electronics devices (IGBT, MOSFET, etc.), a high-speed central controlling unit (a microcontroller, like the PIC18 or the PIC16) and optional sensing devices, depending upon the application requirement.

A typical modern-age intelligent VFD for the three-phase induction motor with single-phase supply is shown in Figure 20.

FIGURE 20: TYPICAL VFD



Note: The presence of particular component(s) and location(s) will depend on the features provided and the technology used in the specific VFD by the manufacturer.

The basic function of the VFD is to act as a variable frequency generator in order to vary speed of the motor as per the user setting. The rectifier and the filter convert the AC input to DC with negligible ripple. The inverter, under the control of the PICmicro® microcontroller, synthesizes the DC into three-phase variable voltage, variable frequency AC. Additional features can be provided, like the DC bus voltage sensing, OV and UV trip, overcurrent protection, accurate speed/position control, temperature control, easy control setting, display, PC connectivity for real-time monitoring, Power Factor Correction (PFC) and so on. With the rich feature set of the PICmicro microcontroller, it is possible to integrate all the features necessary into a single chip solution so as to get advantages, such as reliability, accurate control, space saving, cost saving and so on.

The base speed of the motor is proportional to supply frequency and is inversely proportional to the number of stator poles. The number of poles cannot be changed once the motor is constructed. So, by changing the supply frequency, the motor speed can be changed. But when the supply frequency is reduced, the equivalent impedance of electric circuit reduces. This results in higher current drawn by the motor and a higher flux. If the supply voltage is not reduced, the magnetic field may reach the saturation level. Therefore, in order to keep the magnetic flux within working range, both the supply voltage and the frequency are changed in a constant ratio. Since the torque produced by the motor is proportional to the magnetic field in the air gap, the torque remains more or less constant throughout the operating range.

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SUMMARY

The AC induction motor drive is the fastest growing segment of the motor control market. There are various reasons for this growth. They are:

- Ease of programming
- Low investment cost for development
- Flexibility to add additional features with minimal increase in hardware cost
- Faster time to market
- Same VFD for wide ranges of motors with different ratings
- Reduced total part count and hence, compact design
- Reliability of the end product
- Ease of mass production
- Ever decreasing cost of semiconductors due to advancement in fabrication technology
- Energy efficient solution

Microchip has positioned itself to target the motor control market, where our advanced designs, progressive process technology and industry leading product performance enables us to deliver decidedly superior performance over our competitors, which includes the best of the industry. These products are positioned to provide a complete product solution for embedded control applications found throughout the consumer, automotive and industrial control markets. Microchip products are meeting the unique design requirements of the motion control embedded applications.

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Slip

The difference between the synchronous speed N_s and the actual speed N of the rotor is known as ***slip***. Though it may be expressed in so many revolutions/second, yet it is usual to express it as a percentage of the synchronous speed. Actually, the term '***slip***' is descriptive of the way in which the rotor 'slips back' from synchronism.

$$\% \text{ slip} \quad s = \frac{N_s - N}{N_s} \times 100$$

Sometimes, $N_s - N$ is called the ***slip speed.***

Obviously, rotor (or motor) speed is $N = N_s(1 - s)$.

It may be kept in mind that revolving flux is rotating synchronously, relative to the stator (*i.e.* stationary space) but at slip speed relative to the rotor.

Frequency of Rotor Current

When the rotor is stationary, the frequency of rotor current is *the same as the supply frequency.* But when the rotor starts revolving, then the frequency depends upon the relative speed or on slip-speed. Let at any slip-speed, the frequency of the rotor current be f'. Then

$$N_s - N = \frac{120\,f'}{P} \quad \text{Also,} \quad N_s = \frac{120\,f}{P}$$

Dividing one by the other, we get, $\dfrac{f'}{f} = \dfrac{N_s - N}{N_s} = s$ $\qquad \therefore \quad f' = sf$

Example 34.2. *The stator of a 3-ϕ induction motor has 3 slots per pole per phase. If supply frequency is 50 Hz, calculate*

(i) number of stator poles produced and total number of slots on the stator

(ii) speed of the rotating stator flux (or magnetic field).

Solution. *(i)* $P = 2n = 2 \times 3 =$ **6 poles**

Total No. of slots = 3 slots/pole/phase × 6 poles × 3 phases = **54**

(ii) $N_s = 120\,f/P = 120 \times 50/6 =$ **1000 r.p.m.**

Example 34.3. *A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50 Hz. Calculate :*

(i) the speed at which the magnetic field of the stator is rotating.

(ii) the speed of the rotor when the slip is 0.04.

(iii) the frequency of the rotor currents when the slip is 0.03.

(iv) the frequency of the rotor currents at standstill.

(Electrical Machinery II, Banglore Univ. 1991)

Solution. *(i)* Stator field revolves at synchronous speed, given by

$N_s = 120\,f/P = 120 \times 50/4 =$ **1500 r.p.m.**

(ii) rotor (or motor) speed, $N = N_s(1 - s) = 1500(1 - 0.04) =$ **1440 r.p.m.**

(iii) frequency of rotor current, $f' = sf = 0.03 \times 50 = 1.5$ r.p.s = **90 r.p.m**

(iv) Since at standstill, $s = 1$, $f' = sf = 1 \times f = f =$ **50Hz**

Example 34.4. *A 3-ϕ induction motor is wound for 4 poles and is supplied from 50-Hz system. Calculate (i) the synchronous speed (ii) the rotor speed, when slip is 4% and (iii) rotor frequency when rotor runs at 600 rpm.* **(Electrical Engineering-I, Pune Univ. 1991)**

Solution. *(i)* $N_s = 120\,f/P = 120 \times 50/4 = 1500$ rpm

(ii) rotor speed, $N = N_s(1 - s) = 1500\,(1 - 0.04) =$ **1440 rpm**

(iii) when rotor speed is 600 rpm, slip is

$s = (Ns - N)/N_s = (1500 - 600)/1500 = 0.6$

rotor current frequency, $f' = sf = 0.6 \times 50 =$ **30 Hz**

Example 34.5. *A 12-pole, 3-phase alternator driven at a speed of 500 r.p.m. supplies power to an 8-pole, 3-phase induction motor. If the slip of the motor, at full-load is 3%, calculate the full-load speed of the motor.*

Solution. Let N = actual motor speed; Supply frequency, $f = 12 \times 500/120 = 50$ Hz. Synchronous speed $N_s = 120 \times 50/8 = 750$ r.p.m.

$$\% \text{ slip} \quad s = \frac{N_S - N}{N} \times 100\,; \quad 3 = \frac{750 - N}{750} \times 100 \qquad \therefore N = \textbf{727.5 r.p.m.}$$

Note. Since slip is 3%, actual speed N is less than N_s by 3% of N_s *i.e.* by $3 \times 750/100 = 22.5$ r.p.m.